



Modelling Financial Markets

Chapter 1: Stylised Facts and Returns



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ACF: AutoCorrelation Function
- AI: Artificial Intelligence
- ARCH: AutoRegressive Conditional Heteroskedasticity
- ARCH-LM: ARCH Lagrange Multiplier (test)
- BET: Bucharest Exchange Trading (index)
- BET-TR: BET Total Return (index)
- BNR: Banca Națională a României (the National Bank of Romania)
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CAGR: Compound Annual Growth Rate
- CI: Confidence Interval
- CLT: Central Limit Theorem
- CRIX: CRypto IndeX
- EGARCH: Exponential GARCH
- ETF: Exchange-Traded Fund
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GEO: Government Emergency Ordinance
- GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- JB: Jarque–Bera (test)
- LB: Ljung–Box (test)
- MDD: Maximum DrawDown
- OHLC: Open, High, Low, Close
- QQ: Quantile–Quantile (plot)
- SE: Standard Error
- US: United States

Outline

Motivation

Prices and Returns

Descriptive Statistics and Tests

Stylised Facts

Markets Compared

Volatility Estimators

Risk-Adjusted Performance

Implications for Modelling

AI for Scientific Discovery

Conclusions

References

What Is a Stylised Fact?

Definition

- ▣ A **stylised fact** is a statistical regularity observed across many assets, markets and periods
- ▣ It is qualitative, not quantitative: “returns have heavy tails”, not “kurtosis = 7.3”
- ▣ It describes **returns**, not price levels; the standard survey is [Cont \(2001\)](#)

Key message of this chapter

- ▣ Returns show little **linear autocorrelation**, but their **size** (absolute value, a proxy for risk) is highly structured

Why Stylised Facts Matter

- They are the **benchmark** every model of returns must reproduce
- A model that violates them misprices risk: margins, capital and option prices are wrong
- The textbook model of normally distributed, i.i.d. returns fails almost all of them
 - ▶ i.i.d. = **independent and identically distributed**: every day is a new, independent draw from the same distribution
- Stylised facts tell us **which** features a better model needs
- Later chapters build on them: volatility (5, 6, 9), risk (7, 8), machine learning and bubbles (13, 17)

Data Used in This Chapter

Series	Market	Period
S&P 500 index	US equities (price index)	1990–2026
BET-TR	Romanian equities (total return)	2014–2026
Bitcoin	crypto-asset, traded 24/7	2014–2026
EUR/RON	exchange rate, managed float	Jul 2005–2026
Gold	spot price, USD per ounce	1990–2026

- Daily closing values up to 18 September 2026 (repository folder data/market)
- Unless stated otherwise, returns are daily log returns (defined in Section 2)

The Five Series

- ▣ **S&P 500**: Standard & Poor's index of 500 large US companies; a **price index**, dividends are not included
- ▣ **BVB**: the Romanian stock exchange
 - ▶ **BET**: the benchmark index of the BVB
 - ▶ **BET-TR**: its version with dividends re-invested
- ▣ **EUR/RON**: price of one euro in Romanian lei; a **managed float** (market rate with central-bank interventions)
 - ▶ Source: the daily reference rate of the **BNR**
- ▣ **Gold**: spot price in US dollars (USD) per troy ounce; weekday quotes only (weekend prints removed)
- ▣ **Bitcoin**: price in USD; trades 24 hours a day, 7 days a week (24/7)

Data Quality Notes

- EUR/RON: official BNR reference rate instead of exchange quotes
 - ▶ Exchange quote series showed lag-1 autocorrelation (correlation of consecutive daily changes) of -0.4 to -0.7 in several years
 - ▶ Such values come from quote errors, not from the market
- S&P 500 **OHLC** data: Open, High, Low and Close prices of each day
 - ▶ Open prices before 2007 equal the previous close: unusable for range-based estimators
 - ▶ Range estimators therefore start in 2008 (Section 6)
- Lesson: always inspect the data before computing statistics

Prices Before Computers: the Stock Ticker



Stock ticker designed by Thomas Edison
(Smithsonian collection)

Photo: Jaclyn Nash, Smithsonian National Museum of
American History; public domain; Wikimedia Commons

- Before computers, trades reached brokers as prices printed on a paper **ticker tape**
- A daily close is one snapshot of this stream of prices; returns are built from such snapshots
- The choice of snapshot (close, fixing, a midnight snapshot) matters, as Chapter 0 showed
- Prices at the level of individual trades: market microstructure, Chapter 10

Prerequisites Recap: Returns

	Simple return R_t	Log return r_t
Definition	$(P_t + D_t)/P_{t-1} - 1$	$\ln(1 + R_t) = \ln P_t - \ln P_{t-1}$ (no dividend)
Over k periods	$\prod_j (1 + R_{t-j}) - 1$	$\sum_j r_{t-j}$: additive over time
Across assets	$R_{p,t} = \sum_j w_j R_{i,t}$: exact	only approximately
Range	$R_t \geq -100\%$	$(-\infty, +\infty)$
Use	portfolios, performance reporting	single-asset dynamics, CLT arguments

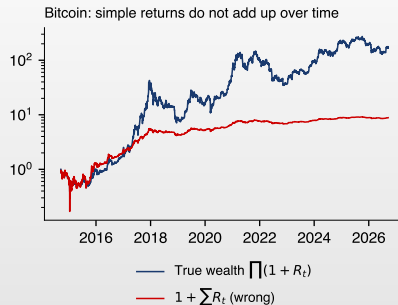
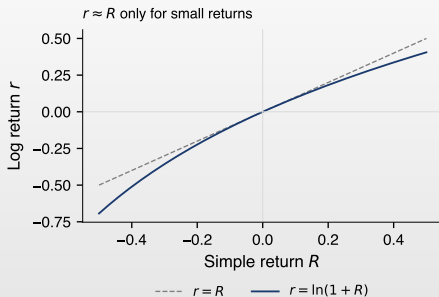
- ▣ Returns, not prices: returns are scale-free; log prices have a unit root, returns are close to stationary
- ▣ Dividends and splits: use adjusted or total-return series (BET-TR); P_t is the close, D_t the dividend paid in $(t-1, t]$
- ▣ $\ln(1 + R) < R$ for every $R \neq 0$: $+1\% \rightarrow 0.995\%$, but $-50\% \rightarrow -69.3\%$
- ▣ Textbooks: [Campbell, Lo & MacKinlay \(1997\)](#), ch. 1; [Franke, Härdle & Hafner \(2019\)](#)

Prerequisites Recap: Annualisation, CAGR and Volatility Drag

- P = periods per year: 252 (stocks), 365 (crypto, traded every day), the actual ≈ 260 weekday quotes (gold)
- $\mu_{\text{ann}} = P \bar{r}$ needs only a constant mean; $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$ is exact for serially uncorrelated returns with constant variance
 - ▶ Weakly stationary returns (constant mean and variance, autocovariances depending only on the lag):

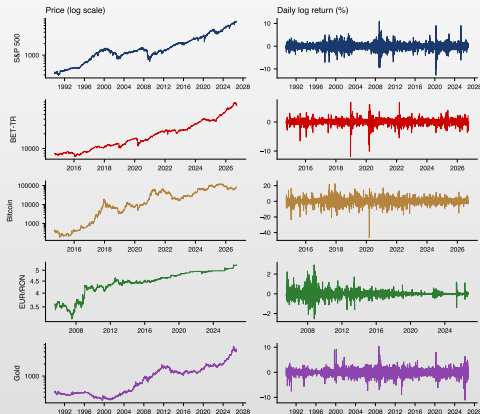
$$\text{Var}\left(\sum_{j=1}^h r_{t+j}\right) = h\sigma^2 \left[1 + 2 \sum_{j=1}^{h-1} (1 - j/h) \rho_j\right]$$
 - ▶ Autoregression of order 1 with $\rho_1 = 0.17$ (EUR/RON): factor $\sqrt{(1 + \rho_1)/(1 - \rho_1)} = 1.19$ for large h , annual volatility 4.4% $\rightarrow$ 5.3%; the variance-ratio test is in Chapter 2
- **CAGR** (Compound Annual Growth Rate) $= (P_T/P_0)^{1/Y} - 1$, Y = number of years; the geometric mean $\approx \mu_{\text{arith}} - \sigma^2/2$ (**volatility drag**)
 - ▶ S&P 500 1990–2026: $\hat{\sigma} = 1.133\%$ a day, 18.0% a year; CAGR 8.69% (price index, no dividends)
 - ▶ Bitcoin 2014–2026: arithmetic mean 65.2% a year ($P = 365$), CAGR 53.9%: a drag of 11 pp (percentage points)

Simple vs Log Returns in Practice



- Left: $r \approx R$ only for small moves; the gap grows with $|R|$
- Right: Bitcoin 2014–2026 multiplied wealth by 176.9, i.e. $\sum_t r_t = \ln 176.9 = 5.18$
- Adding daily simple returns gives $1 + \sum_t R_t = 8.83$: wrong by a factor of 20

Five Markets: Prices and Returns



- ▣ **Left:** prices on log scale, trending and non-stationary
- ▣ **Right:** daily log returns
 - ▶ Mean-reverting around zero
 - ▶ Calm and turbulent periods alternate
 - ▶ Extreme days cluster: 2008, March 2020
- ▣ EUR/RON moves more than 1% on only 1.4% of days
- ▣ Bitcoin, 12 March 2020: log return -46.5% , i.e. a price fall of $1 - e^{-0.465} = 37.2\%$

Recap: Prices and Returns

- Work with returns: scale-free and close to stationary
- Simple returns aggregate across assets; log returns aggregate over time
- Annualise with P and $\sqrt{P}$; use 365 for crypto
- CAGR (geometric) is below the arithmetic mean by about $\sigma^2/2$
- Use total-return or adjusted series when dividends matter

Sample Moments and Their Sampling Error

- ▣ $z_t = (r_t - \bar{r})/\hat{\sigma}$; skewness $\hat{S} = \frac{1}{T} \sum_t z_t^3$, kurtosis $\hat{K} = \frac{1}{T} \sum_t z_t^4$; Normal distribution: $S = 0$, $K = 3$, excess kurtosis $K - 3 = 0$
- ▣ Mean: $SE(\bar{r}) = \sigma/\sqrt{T}$; daily means are 10–60 times smaller than $\hat{\sigma}$, so the mean is the least precise moment
 - ▶ S&P 500: $\bar{r} = 0.033\% \pm 0.023\%$ a day (95%), $T = 9,245$
- ▣ i.i.d. data from the Normal distribution: $\sqrt{T} \hat{S} \rightarrow N(0, 6)$ and $\sqrt{T}(\hat{K} - 3) \rightarrow N(0, 24)$
- ▣ Any i.i.d. distribution: the delta method gives a Normal limit for $\hat{S}$ only if $\mathbb{E}r^6 < \infty$, for $\hat{K}$ only if $\mathbb{E}r^8 < \infty$
- ▣ **Tail index** α : $P(|X| > x) \approx c x^{-\alpha}$ for large x ; $\mathbb{E}|X|^p < \infty$ only for $p < \alpha$
 - ▶ With $\alpha \approx 3$ (Hill estimates, Section 4) $\hat{S}$ has no Normal limit and the population kurtosis does not exist

Three Classical Tests and What They Assume

Test	Statistic	H_0	χ^2 limit needs	Problem here
JB, Jarque & Bera (1980)	$T \left[\frac{\hat{S}^2}{6} + \frac{(\hat{K}-3)^2}{24} \right]$	i.i.d., Normal distribution	H_0 itself (all moments exist)	a rejection does not say which part fails
LB, Ljung & Box (1978)	$T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}$	$\rho_1 = \dots = \rho_m = 0$	i.i.d., not only uncorrelated	clustering: over-rejects
ARCH-LM, Engle (1982)	$TR^2, \hat{\varepsilon}_t^2$ on q lags	i.i.d., no ARCH effects	$\mathbb{E}r^4 < \infty$	$\alpha \approx 3 < 4$: size unreliable

- ▣ **ARCH** = AutoRegressive Conditional Heteroskedasticity: today's variance depends on past returns; $\hat{\varepsilon}_t = r_t - \bar{r}$, $q = 5$
- ▣ We use $m = 10$; 5% critical values: $\chi_2^2 = 5.99$, $\chi_5^2 = 11.1$, $\chi_{10}^2 = 18.3$
- ▣ The question is not how to run the tests but whether their reference distribution holds for returns

Worked Example: JB for the S&P 500

- ▣ $JB = T \left[\hat{S}^2/6 + (\hat{K} - 3)^2/24 \right]$
- ▣ S&P 500, 1990–2026: $T = 9,245$, $\hat{S} = -0.364$, $\hat{K} - 3 = 10.88$
- ▣ Skewness term: $\hat{S}^2/6 = 0.132/6 = 0.022$
- ▣ Kurtosis term: $10.88^2/24 = 118.4/24 = 4.93$
- ▣ $JB = 9,245 \times (0.022 + 4.93) \approx 45,800$, against the critical value 5.99
- ▣ Almost all of the statistic comes from the kurtosis term: the problem is the tails, not the asymmetry

What JB Tests, and What a Rejection Means

- ▣ JB is a Lagrange Multiplier test [Jarque & Bera \(1980\)](#): under i.i.d. returns from the Normal distribution $\sqrt{T}(\hat{S}, \hat{K} - 3) \rightarrow N(0, \text{diag}(6, 24))$, hence χ^2_2
- ▣ Under this null all moments exist: the p-value is valid for the **joint** null “i.i.d. and Normal distribution”
 - ▶ Heavy tails (Hill $\hat{\alpha} \approx 3$) make JB grow with T : this is the power of the test, not a broken p-value
- ▣ A rejection does not say which part fails: the marginal distribution or independence (GARCH dependence changes $\text{Var}(\hat{S})$ and $\text{Var}(\hat{K})$)
 - ▶ [Bai & Ng \(2005\)](#): tests of skewness, kurtosis and normality of the marginal distribution of a time series, studentised with HAC variances
 - ▶ Their separate skewness and kurtosis tests (null $S = 0$ or $K = 3$ without normality) need $\mathbb{E}r^6 < \infty$ and $\mathbb{E}r^8 < \infty$
- ▣ Report as well: quantile-based skewness and kurtosis (next slide) and the tail index (Section 4)

Robust Alternative: Quantile-Based Skewness and Kurtosis

- Hinkley skewness $\frac{q_{0.95} + q_{0.05} - 2q_{0.5}}{q_{0.95} - q_{0.05}}$; Crow–Siddiqui kurtosis $\frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$; both 0 for the Normal distribution [Kim & White \(2004\)](#)
- No moments needed, only positive denominators (interquantile ranges); 95% CI from a moving-block bootstrap (blocks of 20 days, 1,000 resamples), valid for stationary, weakly dependent returns with a positive density at these quantiles

	Moment skew.	Hinkley [95% CI]	Excess kurt.	Crow–Siddiqui [95% CI]
S&P 500	−0.36	−0.064 [−0.092, −0.041]	10.9	1.54 [1.29, 1.79]
BET-TR	−1.40	−0.036 [−0.085, 0.025]	17.8	1.06 [0.73, 1.43]
Bitcoin	−0.70	−0.018 [−0.069, 0.032]	12.0	2.25 [1.88, 2.64]
EUR/RON	0.73	0.041 [−0.022, 0.093]	15.1	4.87 [3.74, 6.13]
Gold	−0.23	0.004 [−0.025, 0.030]	9.3	1.47 [1.29, 1.73]

- Heavy tails survive in all five markets: no CI contains 0
- Negative skewness survives only for the S&P 500: BET-TR's −1.40 comes from a few days (19 December 2018)

The ACF Under Conditional Heteroskedasticity

- ▣ **ACF** (AutoCorrelation Function): $\hat{\rho}_k = \sum_t x_t x_{t-k} / \sum_t x_t^2$, $x_t = r_t - \bar{r}$, plotted against k
- ▣ Martingale difference (uncorrelated, but $\mathbb{E}[r_t^2 \mid \text{past}]$ varies): $\hat{\rho}_k \approx N(0, \tau_k)$, $\hat{\tau}_k = \sum_t x_t^2 x_{t-k}^2 / (\sum_t x_t^2)^2$ [Romano & Thoms \(1996\)](#)
 - ▶ i.i.d.: $\tau_k = 1/T$ and the band $\pm 1.96/\sqrt{T}$; volatility clustering: $T\tau_k > 1$, the i.i.d. band is too narrow

Lag 1	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
$\hat{\rho}_1$	-0.079	0.061	-0.022	0.170	-0.011
$T\hat{\tau}_1$	5.00	4.35	2.89	6.85	2.43
t-statistic, i.i.d. / robust	-7.6 / -3.4	3.3 / 1.6	-1.5 / -0.9	12.4 / 4.7	-1.1 / -0.7

- ▣ Robust portmanteau $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k$ [Lobato, Nankervis & Savin \(2001\)](#): S&P 500 $Q(10) = 91.6$ but $\tilde{Q}(10) = 19.0$ ($p = 0.041$)
 - ▶ Needs $\mathbb{E}r^4 < \infty$ and $\mathbb{E}[x_t^2 x_{t-k} x_{t-l}] = 0$ for $k \neq l$; [Dalla, Giraitis & Phillips \(2022\)](#) keep only significant cross terms: none for the S&P 500
- ▣ ACF of $|r_t|$ and r_t^2 : Normal limits need $\alpha > 4$ and $\alpha > 8$ ([Davis & Mikosch \(1998\)](#), Section 4)

Summary Statistics: Five Markets

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Observations	9245	2992	4384	5352	9559
Mean (% per day)	0.033	0.077	0.118	0.007	0.025
Std. dev. (% per day)	1.13	0.98	3.49	0.28	1.01
Annualised volatility (%)	18.0	15.5	66.6	4.4	16.3
Skewness	-0.36	-1.40	-0.70	0.73	-0.23
Excess kurtosis	10.9	17.8	12.0	15.1	9.3
Min / max log return (%)	-12.8 / 11.0	-11.9 / 6.7	-46.5 / 22.5	-2.5 / 2.9	-11.1 / 10.4
Jarque-Bera	45805	40433	26482	51575	34732
Ljung-Box $Q(10), r_t$	91.6	70.0	20.2	221.7	32.1
Ljung-Box $Q(10), r_t^2$	8018	714	213	2420	1049
ARCH-LM (5 lags)	2229	371	115	871	483

- Min and max are daily **log** returns; EUR/RON from 1 July 2005 (new leu, RON)
- 5% critical values: $\chi_2^2 = 5.99$, $\chi_{10}^2 = 18.3$, $\chi_5^2 = 11.1$; all i.i.d. p-values < 0.03, valid only under the assumptions above

Reading the Table

- ▣ **“i.i.d. and Normal distribution” is rejected everywhere:** JB in the tens of thousands; the quantile-based kurtosis shows that the tails are part of the reason
- ▣ **Heavy tails:** excess kurtosis between 9.3 (gold) and 17.8 (BET-TR)
- ▣ **Negative skewness** for equity, crypto and gold; strongest for BET-TR (-1.40)
- ▣ **Volatility is predictable:** $Q(10)$ on r_t^2 is 10–90 times larger than on r_t ; ARCH-LM rejects everywhere
- ▣ **Linear autocorrelation** of r_t : significant with i.i.d. tests, economically small
 - ▶ With robust standard errors only the S&P 500 (lag 1, $t = -3.4$) and EUR/RON ($t = 4.7$) remain significant
- ▣ EUR/RON is special: tiny volatility, high excess kurtosis, positive skewness, strong autocorrelation (Section 5)

Recap: Descriptive Statistics and Tests

- ▣ Skewness and kurtosis have Normal limits only if $\mathbb{E}r^6$ and $\mathbb{E}r^8$ are finite
- ▣ JB tests the joint null “i.i.d. and Normal distribution”; Ljung–Box and ARCH-LM have χ^2 limits only under i.i.d.-type assumptions that returns violate
- ▣ Robust versions: quantile-based moments, heteroskedasticity-robust ACF bands and $\tilde{Q}$
- ▣ Heavy tails survive every robust check; linear predictability mostly does not
- ▣ Statistical significance is not economic significance in samples of thousands of days

The People Behind the Stylised Facts (1/2)



Carl Friedrich Gauss (1777–1855)

Painting: C.A. Jensen (1840); public domain;
Wikimedia Commons

- The Normal distribution: the benchmark that returns violate



Louis Bachelier (1870–1946)

Photo: unknown author; public domain;
Wikimedia Commons

- Prices as a random walk, in his thesis Bachelier (1900)

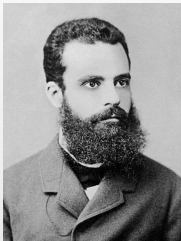


William S. Gosset, “Student” (1876–1937)

Photo: unknown author; public domain;
Wikimedia Commons

- The Student- t distribution Student (1908), today a model for heavy tails

The People Behind the Stylised Facts (2/2)



Vilfredo Pareto (1848–1923)

Photo: unknown author (1870s); public domain; [Wikimedia Commons](#)



Benoît Mandelbrot (1924–2010)

Photo: Rama (2007); CC BY-SA 2.0 France; [Wikimedia Commons](#)

- Pareto: **power laws** for large incomes, the ancestor of heavy-tail models
- Mandelbrot: cotton prices have heavy tails and the Normal-distribution model fails [Mandelbrot \(1963\)](#)
- Their tail index α is what the Hill estimator in this chapter measures

The Stylised Facts of Cont (2001)

1. Absence of linear autocorrelation
 2. Heavy tails
 3. Gain/loss asymmetry
 4. Aggregational Gaussianity
 5. Intermittency
 6. Volatility clustering
 7. Conditional heavy tails
 8. Slow decay of autocorrelation in absolute returns
 9. Leverage effect
 10. Volume/volatility correlation
 11. Asymmetry in time scales
- ▣ We verify most of them on the five markets, with data up to September 2026
 - ▣ Early evidence: Mandelbrot (1963), Fama (1965); textbook treatment: Taylor (2008), Franke, Härdle & Hafner (2019)

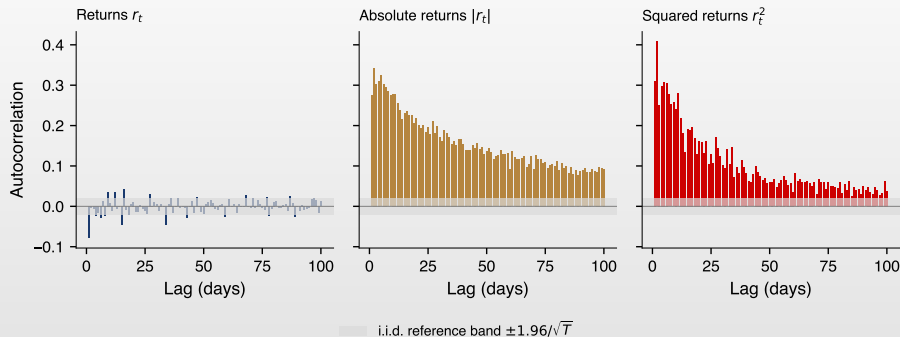
Do Cont's Facts Still Hold? Recent Evidence

- ▣ Ratliff-Crain, Van Oort, Koehler & Tivnan (2025)
 - ▶ Re-test all 11 facts on intraday returns of the Dow 30 stocks, October 2018 – March 2019
 - ▶ Conclusive evidence for 8 facts, no support for the remaining 3: a fact is a hypothesis to re-test on each market and frequency
- ▣ Heavy tails beyond the Student- t : estimation of alpha-stable distributions, whose tail index is below 2 [Pele \(2014\)](#)
- ▣ Entropy-based measures of tail risk as an alternative to kurtosis [Pele, Lazar & Dufour \(2017\)](#)
- ▣ Crypto-assets: the same facts, with a tail factor that separates them from other asset classes [Pele et al. \(2023\)](#)
- ▣ This chapter re-tests the facts on five markets and adds the inference that each test needs

Facts 1 and 6: Uncorrelated but Not Independent

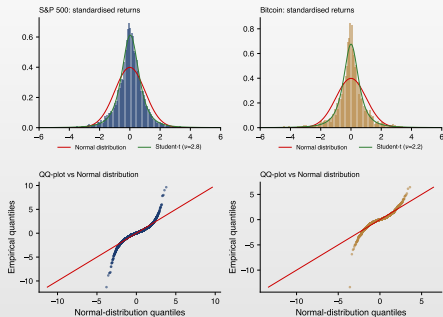
- **Fact 1 — absence of autocorrelation:** $\text{corr}(r_t, r_{t-k}) \approx 0$ for $k \geq 1$
 - ▶ Past returns do not predict the direction of future returns linearly
- **Fact 6 — volatility clustering:** large moves tend to be followed by large moves, of either sign
 - ▶ $\text{corr}(|r_t|, |r_{t-k}|) > 0$ for many lags
- Together: returns are **uncorrelated but not independent**
- Independence would require zero correlation of **every** function of returns, including $|r_t|$
- Modelling: $r_t = \sigma_t \varepsilon_t$ with σ_t driven by past returns (GARCH, Chapter 5); linear autoregressive forecasts of r_t add little

Facts 1 and 6 on the S&P 500



- S&P 500 1990–2026, lags 1–100; grey band: i.i.d. $\pm 1.96/\sqrt{T}$; the robust band is 2.2 times wider at lag 1
- Returns: almost all autocorrelations inside the band (lag 1: -0.08)
- Absolute and squared returns: large, positive and persistent (lag 1: 0.27 for $|r_t|$)

Heavy Tails: Histograms and QQ-Plots



- Fact 2: extreme returns are far more frequent than under a Normal distribution with the same variance
- Standardised returns $(r_t - \bar{r})/\hat{\sigma}$: higher peak and fatter tails than the Normal distribution
- A Student- t fits much better (next slides); QQ-plots (empirical quantiles against those of the Normal distribution) bend away from the line in both tails

How Heavy Are the Tails?

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Days beyond $3\hat{\sigma}$ (%)	1.55	1.44	1.82	2.06	1.42
Days beyond $5\hat{\sigma}$ (count)	30	12	10	29	26
Expected under the Normal distribution, $> 5\hat{\sigma}$	0.005	0.002	0.003	0.003	0.005
Student- t degrees of freedom $\hat{\nu}$ (maximum likelihood)	2.8	3.1	2.2	1.2	2.9

- Normal distribution: 0.27% of days beyond 3σ ; observed 5–8 times more
- Normal distribution: one 5σ day every 6,900 years of trading; observed 10–30 in 12–36 years
- $\hat{\nu}$: Student- t degrees of freedom (next slide), estimated by **maximum likelihood**: the value that makes the observed returns most probable

The Student- t Distribution: Tails, Moments and Our Fits

- Density with ν degrees of freedom, location μ , scale s : $f(x) \propto \left(1 + \frac{(x-\mu)^2}{\nu s^2}\right)^{-(\nu+1)/2}$; $\nu \rightarrow \infty$: the Normal distribution; $\nu = 1$: Cauchy; for returns: [Praetz \(1972\)](#)
- Power-law tail $P(X > x) \propto x^{-\nu}$: moments exist only of order $< \nu$
 - ▶ Variance $s^2\nu/(\nu - 2)$ for $\nu > 2$; excess kurtosis $6/(\nu - 4)$ for $\nu > 4$
- Our fits $\hat{\nu} \approx 1.2$ – 3.1 are **model-based** tail indices: valid if the Student- t holds everywhere; the many small returns in the centre dominate the likelihood, so a misfit centre moves $\hat{\nu}$
- $\hat{\nu} \leq 2$ (EUR/RON: 1.2) implies infinite variance; a finite sample always has a finite variance, so this alone is no contradiction
 - ▶ But Hill (tail only) gives 2.8: the disagreement signals misspecification (the fixing: 57% of daily changes below 0.1%, plus rare jumps)
- The tail itself: the Hill estimator (next slides); power laws were first argued by [Mandelbrot \(1963\)](#) and [Fama \(1965\)](#)

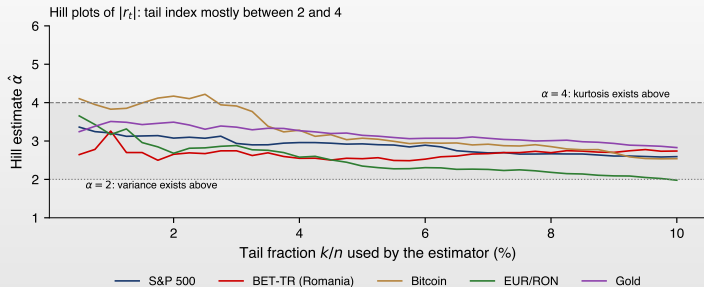
The Hill Estimator of the Tail Index

- ▣ **Tail index** α : $P(|X| > x) \approx c x^{-\alpha}$ for large x ; moments of order $< \alpha$ exist
 - ▶ $\alpha > 2$: finite variance; $\alpha > 4$: finite kurtosis
- ▣ **Hill estimator**, [Hill \(1975\)](#): order the absolute returns $X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(n)}$ and use the k largest

$$\hat{\alpha}_k = \left[\frac{1}{k} \sum_{i=1}^k \ln X_{(i)} - \ln X_{(k+1)} \right]^{-1}$$

- ▣ **Choice of k** : a bias–variance trade-off
 - ▶ Small k : few observations, noisy; large k : the centre of the distribution enters, biased
 - ▶ **Hill plot**: $\hat{\alpha}_k$ against k ; read the level where the curve is flat (here $k = 1\text{--}5\%$ of n)

Hill Plots for Five Markets



- $\hat{\alpha}$ for $|r_t|$ at $k = 2.5\%$ of n : S&P 500 3.1, BET-TR 2.7, Bitcoin 4.2, EUR/RON 2.8, gold 3.3
- For $k = 1\text{--}5\%$: $\hat{\alpha}$ between 2.3 and 3.5 except Bitcoin (3.1–4.2); loss tails heavier for BET-TR (2.2 vs 3.1) and Bitcoin (2.7 vs 3.4)
- Variance exists, but the **fourth moment very likely does not**: sample kurtosis never settles (confidence intervals: next slides); tails are modelled with extreme value theory (Chapter 7)

Hill Estimator: Asymptotics and the Choice of k

- ▣ Hill = maximum likelihood estimator of a Pareto tail above $u = X_{(k+1)}$, conditional on u (Seminar A11)
- ▣ If $k \rightarrow \infty$ and $k/n \rightarrow 0$, and the tail is Pareto up to a second-order term

$$\sqrt{k}(\hat{\alpha}_k - \alpha) \rightarrow N(0, \alpha^2), \quad \text{SE}(\hat{\alpha}_k) \approx \hat{\alpha}_k / \sqrt{k}$$

- ▶ If k grows too fast, a bias term appears: the tail is only approximately Pareto
- ▣ Dependent data (GARCH-type): $\hat{\alpha}$ stays consistent, but extremes cluster and the variance exceeds α^2/k [Hsing \(1991\)](#)
- ▣ Choosing k : plot $\hat{\alpha}$ against $\log k$ ("alt-Hill" plot) [Drees, Resnick & de Haan \(2000\)](#)
- ▣ Regression alternative: log rank on log size, with rank shifted by $-\frac{1}{2}$; $\text{SE} = \hat{\alpha}\sqrt{2/k}$ [Gabaix & Ibragimov \(2011\)](#)

Tail Indices with Confidence Intervals

$ r_t $, $k = 2.5\%$ of n	n	k	$\hat{\alpha}$	SE i.i.d.	95% CI (i.i.d.)	SE block bootstrap
S&P 500	9245	231	3.07	0.20	[2.68, 3.47]	0.33
BET-TR	2992	74	2.67	0.31	[2.06, 3.28]	0.45
Bitcoin	4384	109	4.22	0.40	[3.43, 5.01]	0.47
EUR/RON	5352	133	2.82	0.24	[2.34, 3.30]	0.28
Gold	9559	238	3.31	0.21	[2.89, 3.73]	0.24

- Moving-block bootstrap (blocks of 20 days, 500 resamples) keeps the clustering of extremes: SEs 1.1–1.6 times the i.i.d. ones
- $\alpha = 4$ is outside every interval except Bitcoin's, also with the bootstrap SE: the fourth moment very likely does not exist
- $\alpha = 2$ is outside every i.i.d. interval; with the bootstrap SE only BET-TR reaches it ($2.67 - 1.96 \times 0.45 = 1.79$)

Two Tails and the Tail of $|r_t|$

- ▣ $P(|X| > x) = P(X > x) + P(X < -x)$: with indices α_- (losses) and α_+ (gains), $|X|$ has index $\min(\alpha_-, \alpha_+)$
- ▣ At the same k the estimates are not comparable: the k largest $|r_t|$ come from both tails, so their threshold lies further out
- ▣ Common threshold u : the lower of the two one-sided thresholds at $k = 2.5\%$ of n

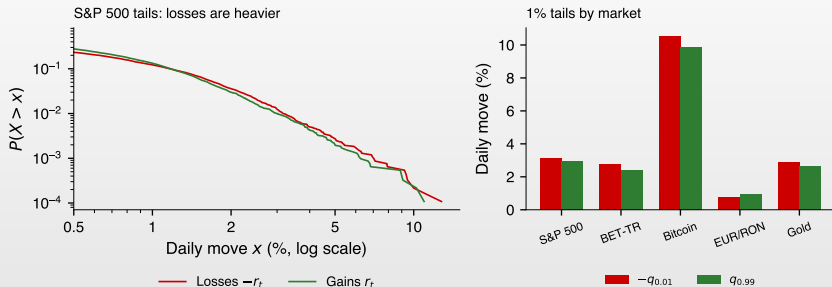
	u (%)	k for $ r_t $	$\hat{\alpha}$, $ r_t $	$\hat{\alpha}_-$, losses	$\hat{\alpha}_+$, gains
S&P 500	2.17	508	2.90	2.85	2.96
BET-TR	1.79	153	2.54	2.20	3.06
Bitcoin	7.11	233	3.02	2.71	3.36
EUR/RON	0.54	295	2.28	2.63	2.04
Gold	1.98	513	3.11	2.97	3.28

- ▣ At a common u , the $|r_t|$ estimate lies between the two tails
- ▣ Bitcoin's 4.2 at $k = 2.5\%$ uses a higher threshold, where its Hill plot is not yet flat: second-order bias; report both one-sided indices

Why Heavy Tails Matter for Modelling

- ▣ **VaR** = Value-at-Risk: the loss exceeded with a small probability (e.g. 1%) over a horizon
 - ▶ It is a quantile of the loss distribution (Chapter 7)
- ▣ A Normal-distribution model with the right variance **underestimates** extreme quantiles
- ▣ Stress scenarios based on “ 3σ events” are far too optimistic
- ▣ Estimators based on high moments (kurtosis) are unstable: prefer quantiles and tail counts
- ▣ Remedies: heavy-tailed distributions (Student- t , possibly skewed), extreme value theory (Chapter 7)

Fact 3: Gain/Loss Asymmetry in the Data

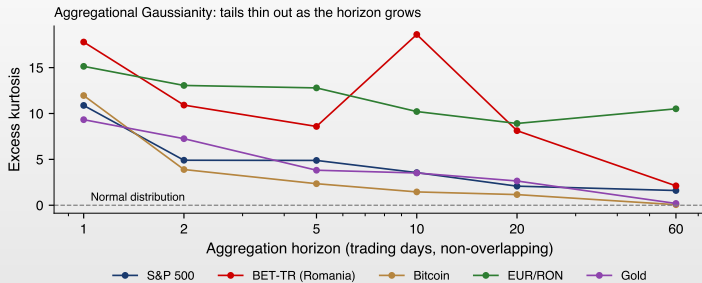


- Fact 3: large losses are more frequent than gains of the same size; survival functions $P(X > x)$ of losses ($-r_t$) and gains (r_t), log-log scale
- VaR 1% ($-q_{0.01}$, the size of the 1% return quantile) exceeds the 99% return quantile $q_{0.99}$ for 4 of 5 markets
 - S&P 500: 3.15% vs 2.97%; BET-TR: 2.73% vs 2.40%; Bitcoin: 10.5% vs 9.9%
- EUR/RON is the exception: leu depreciations (up-moves) are larger (0.94% vs 0.76%)

Fact 4: Aggregational Gaussianity

- **Statement:** as the horizon grows, returns get closer and closer to the Normal distribution
- h -day log return: $r_t(h) = r_t + r_{t-1} + \cdots + r_{t-h+1}$
 - ▶ We use non-overlapping blocks, so that the h -day returns do not share days
- Mechanism: the CLT applies to sums of many daily returns
- Convergence is slow because daily returns are dependent (volatility clustering)

Aggregational Gaussianity in the Data



- ▣ Excess kurtosis of h -day returns, $h = 1, \dots, 60$
- ▣ Tails thin out: S&P 500 from 10.9 (daily) to 1.6 ($h = 60$), Bitcoin from 12.0 to 0.1
- ▣ $h = 60$ is a quarter for the S&P 500 (252 days a year) but two months for Bitcoin (365 days a year)
- ▣ At $h = 60$ only 154 (S&P 500) and 73 (Bitcoin) non-overlapping returns remain: approximate standard error $\sqrt{24/n} = 0.39$ and 0.57 under normality, so these values are very imprecise

Why Heavy Tails? Mixtures of Normal Distributions

- ▣ Suppose $r_t = \sigma_t \varepsilon_t$ with $\varepsilon_t \sim N(0, 1)$ independent of a random volatility σ_t

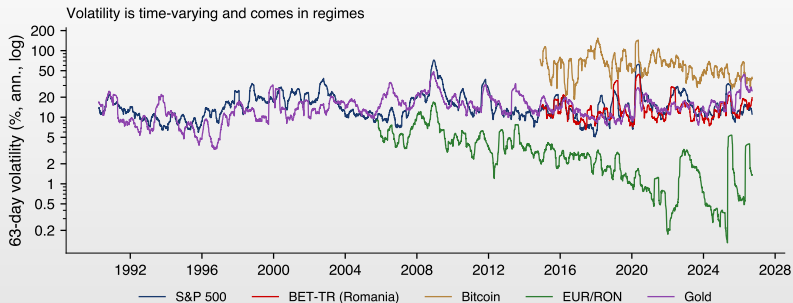
$$K = \frac{\mathbb{E}[r_t^4]}{\mathbb{E}[r_t^2]^2} = 3 \frac{\mathbb{E}[\sigma_t^4]}{\mathbb{E}[\sigma_t^2]^2} \geq 3$$

- ▣ Jensen's inequality ($\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$) with $X = \sigma^2$: any variation in volatility creates excess kurtosis
- ▣ Example: $\sigma = 1\%$ on 90% of days and 3% on 10% of days

$$K = 3 \cdot \frac{0.9 \cdot 1 + 0.1 \cdot 81}{(0.9 + 0.1 \cdot 9)^2} = 3 \cdot \frac{9.0}{3.24} = 8.33$$

- ▣ This links heavy tails (Fact 2) to volatility clustering (Fact 6)

Fact 5: Intermittency in the Data

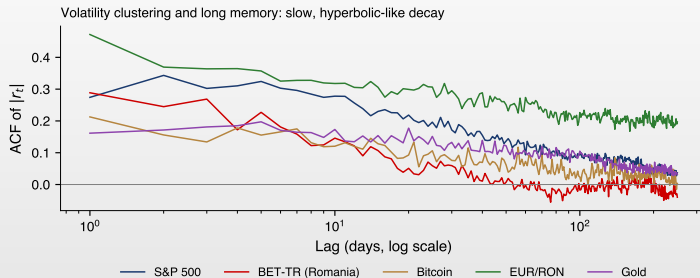


- Fact 5: bursts of high volatility separated by calm periods; constant-volatility models cannot produce them
- 63-day rolling volatility $\sqrt{P} \times$ the standard deviation over the window, log scale
- Common spikes across markets: 2008–2009, March 2020
- EUR/RON volatility collapsed after 2012: an exchange-rate regime managed by the central bank

Fact 8: Slow Decay — Long Memory in Volatility

- **Statement:** the ACF of $|r_t|$ stays positive for months
- **Short memory:** autocorrelations decay **exponentially**, $\rho_k \approx c \phi^k$ with $|\phi| < 1$
 - ▶ Example: GARCH(1,1) volatility (Chapter 5)
- **Long memory:** autocorrelations decay **hyperbolically**, $\rho_k \approx c k^{2d-1}$, memory parameter $0 < d < \frac{1}{2}$
 - ▶ Shocks to volatility die out very slowly
- Evidence: [Ding, Granger & Engle \(1993\)](#)

Long Memory in the Data



- ACF of $|r_t|$ up to one year (250 lags), log scale on the lag axis
- S&P 500: 0.27 at lag 1, 0.22 at lag 20, still 0.09 at lag 100
- BET-TR decays fastest (shorter sample); decay is slower than the exponential decay of GARCH(1,1)

Estimating the Memory Parameter d

- Long memory: spectral density $f(\lambda) \approx G\lambda^{-2d}$ as $\lambda \rightarrow 0$, $0 < d < \frac{1}{2}$; semiparametric estimators use only the first m Fourier frequencies $\lambda_j = 2\pi j/T$ and the periodogram $I(\lambda_j)$
- Log-periodogram regression: $\ln I(\lambda_j)$ on $-2 \ln(2 \sin(\lambda_j/2))$, slope $\hat{d}$, $\text{SE} = \pi/\sqrt{24m}$ Geweke & Porter-Hudak (1983)
- Local Whittle: $\hat{d} = \arg \min_d [\ln(\frac{1}{m} \sum_j \lambda_j^{2d} I(\lambda_j)) - \frac{2d}{m} \sum_j \ln \lambda_j]$, $\sqrt{m}(\hat{d} - d) \rightarrow N(0, \frac{1}{4})$, robust to non-Gaussianity Robinson (1995)

$ r_t $, $m = T^{0.65}$	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
m	378	181	232	265	386
Local Whittle $\hat{d}$ (SE)	0.48 (0.03)	0.37 (0.04)	0.35 (0.03)	0.39 (0.03)	0.40 (0.03)
Log-periodogram $\hat{d}$ (SE)	0.50 (0.03)	0.41 (0.05)	0.35 (0.04)	0.38 (0.04)	0.42 (0.03)

- $\hat{d} \approx 0.35$ – 0.50 everywhere, at least 8 SEs from $d = 0$ (short memory) with both estimators; the S&P 500 is close to the stationarity boundary $\frac{1}{2}$

Long Memory or Structural Breaks?

- Rare shifts in the level of volatility produce a slowly decaying ACF of $|r_t|$ and $\hat{d} > 0$ without long memory Diebold & Inoue (2001)
- Changes in the unconditional variance mimic long memory and integrated GARCH Mikosch & Stărică (2004)
- Check: locate one variance break with the cumulative sum of squares Inclán & Tiao (1994), then re-estimate d (local Whittle) on each side

$ r_t $	Break date	Volatility before / after (%)	$\hat{d}$ before (SE)	$\hat{d}$ after (SE)
S&P 500	20 Jul 1998	12.6 / 19.3	0.28 (0.04)	0.51 (0.03)
BET-TR	17 Dec 2018	12.5 / 16.9	0.19 (0.05)	0.37 (0.04)
Bitcoin	19 Jun 2022	74.9 / 47.7	0.39 (0.04)	0.15 (0.05)
EUR/RON	27 Jun 2013	6.6 / 2.2	0.35 (0.04)	0.35 (0.04)
Gold	29 Nov 2005	13.2 / 18.3	0.30 (0.03)	0.37 (0.03)

- $\hat{d}$ stays well above 0 in every sub-sample: one break does not explain the memory
- But $\hat{d}$ moves (Bitcoin $0.39 \rightarrow 0.15$): several breaks or a time-varying d remain open; the break dates are descriptive

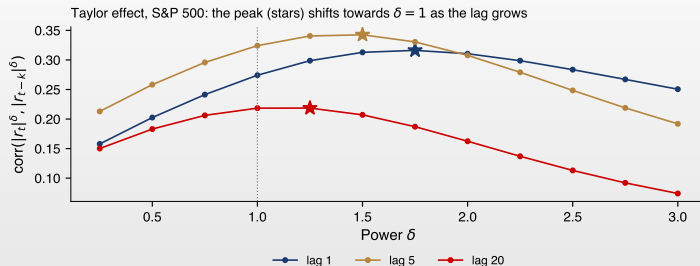
Infinite Fourth Moments and Statistics on r_t^2

- ▣ The ACF of r_t^2 is a ratio of fourth-order moments: it needs $E[r_t^4] < \infty$, i.e. tail index $\alpha > 4$
- ▣ Our Hill estimates are about 2.5–3.5: the fourth moment is very likely infinite
 - ▶ The sample ACF of r_t^2 then converges slowly, to a limit that is not the Normal distribution; the $\pm 1.96/\sqrt{T}$ band is not valid, [Davis & Mikosch \(1998\)](#)
 - ▶ Ljung–Box on r_t^2 and ARCH-LM inherit the problem: the statistics are huge, but their p-values are not reliable
- ▣ The conclusion “volatility clusters” is robust; the exact magnitudes are not
- ▣ Remedy: use $|r_t|$, whose ACF needs only $\alpha > 2$ — this is one reason behind the Taylor effect (next slide)

The Taylor Effect

- **Definition:** compute the autocorrelation of $|r_t|^\delta$ for different powers $\delta > 0$
 - ▶ $\delta = 1$: absolute returns; $\delta = 2$: squared returns
- **Taylor effect:** the autocorrelation is highest for δ around 1, not for $\delta = 2$
- Implication: absolute returns are a more persistent, more robust measure of volatility than squared returns
- Documented by Taylor (1986); the name “Taylor effect” was given by Granger & Ding (1995); see also Ding, Granger & Engle (1993)

The Taylor Effect in the Data



- S&P 500: autocorrelation of $|r_t|^\delta$ as a function of δ
- Maximum at $\delta = 1.75$ (lag 1), 1.5 (lag 5) and 1.25 (lag 20): the optimum moves towards 1 at longer lags
- Motivates power-GARCH models and absolute-value volatility measures

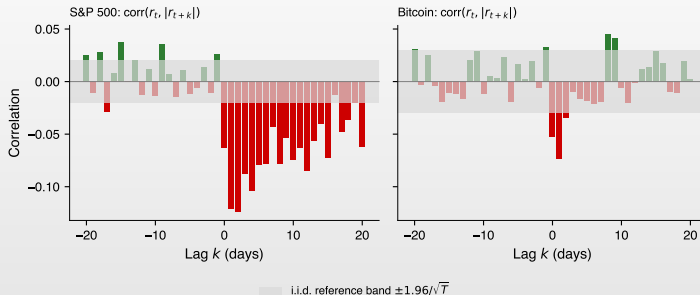
Fact 9: The Leverage Effect

- ▣ **Statement:** falling prices raise **future** volatility more than rising prices
- ▣ **Leverage correlation** at lag k

$$L(k) = \text{corr}(r_t, |r_{t+k}|)$$

- ▶ $L(k) < 0$ for $k > 0$: a negative return today is followed by larger moves
- ▶ $L(k) \approx 0$ for $k < 0$: past volatility does not predict today's sign
- ▣ Explanations: financial leverage of firms, [Christie \(1982\)](#); the retarded volatility model, [Bouchaud, Matacz & Potters \(2001\)](#)

The Leverage Effect in the Data

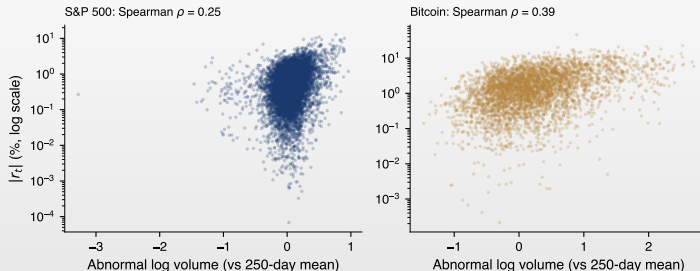


- S&P 500: $L(1) = -0.12$, still -0.07 at $k = 10$; no effect for $k < 0$
- BET-TR -0.12 ; weak for Bitcoin (-0.07), negligible for gold (-0.02), reversed for EUR/RON ($+0.10$)
- Modelling: asymmetric volatility models, GJR-GARCH (Glosten–Jagannathan–Runkle) and EGARCH (Exponential GARCH), Chapter 5

Fact 10: Volume and Volatility Move Together

- ▣ **Statement:** days with large price moves are days with high trading volume
- ▣ **Abnormal log volume:** $\ln V_t$ minus its 250-day moving average
- ▣ Measure: **Spearman rank correlation**, the correlation of ranks, robust to outliers and monotone non-linearity
- ▣ Mixture-of-distributions hypothesis, [Clark \(1973\)](#): prices move in a random “business time” driven by information flow, which also drives volume
- ▣ Survey of the price–volume relation: [Karpoff \(1987\)](#)

Volume and Volatility in the Data



- Abnormal log volume against $|r_t|$
- Spearman rank correlation: 0.25 for the S&P 500 and 0.39 for Bitcoin
- Volume is a useful additional input for volatility forecasts
- Bitcoin caveat: volume is exchange-reported and aggregated over many venues; part of it is **wash trading** (an exchange trading with itself to inflate volume), [Cong, Li, Tang & Yang \(2023\)](#)

Facts 7 and 11: Conditional Tails and Time Scales

Fact 7: conditional heavy tails

- After removing volatility clustering, standardised returns are still heavy-tailed
- Fit a volatility model, e.g. GARCH (Chapter 5), get $\hat{\sigma}_t$, compute $\hat{z}_t = r_t / \hat{\sigma}_t$
- Kurtosis of $\hat{z}_t$: lower than for r_t , still above 3
- Models need heavy-tailed innovations (Student- t , possibly skewed)

□ Both facts are documented in [Cont \(2001\)](#)

□ Returns standardised by realised volatility from intraday data are close to the Normal distribution [Andersen, Bollerslev, Diebold & Ebens \(2001\)](#) (Chapter 9)

Fact 11: asymmetry in time scales

- Weekly volatility predicts daily volatility better than the reverse
- Traders with different horizons react at different frequencies
- **HAR-RV**: Heterogeneous AutoRegressive model of Realised Volatility (Chapter 9)

Recap: The Stylised Facts

- ▣ Uncorrelated returns, strongly dependent volatility (Facts 1, 6, 8)
- ▣ Heavy tails, also conditionally, that thin out slowly with aggregation (Facts 2, 4, 7)
- ▣ Crashes sharper than rallies in equity markets (Facts 3, 9)
- ▣ Volatility in regimes, linked to volume and information flow (Facts 5, 10)
- ▣ Different horizons interact asymmetrically (Fact 11)

Crypto-Assets: The Same Facts, Amplified

Bitcoin vs S&P 500

- ▣ Volatility 66.6% vs 18.0% per year
- ▣ Excess kurtosis 12.0 vs 10.9
- ▣ Skewness -0.70 vs -0.36
- ▣ Worst day, log return: -46.5% (12 March 2020) vs -12.8% (16 March 2020)
- ▣ As price changes: -37.2% vs -12.0%

But some facts are weaker

- ▣ Leverage effect: weak, short-lived
- ▣ Volatility memory: ACF of $|r_t|$ decays faster (lag 100: 0.02 vs 0.09)
- ▣ Trading 24/7: 365 periods per year

Crypto-Assets: An Asset Class of Their Own

- Crypto as an asset class with its own statistical profile: [Härdle, Harvey & Reule \(2020\)](#)
- Tail, memory and moment factors classify cryptos as a separate asset class, mainly through the tail factor: [Pele et al. \(2023\)](#)
- Crypto modelling and the CRIX index in quantitative finance: [Härdle, Chen & Overbeck \(2017\)](#)
 - ▶ CRIX = CRypto IndeX, a benchmark index of the crypto market
- Deeper treatment in Chapter 16 (digital assets)

EUR/RON: A Managed Exchange Rate

- ▣ **Managed float:** the exchange rate is market-determined, but the central bank intervenes to limit fluctuations
- ▣ BNR reference rate, 1 July 2005–2026: from the new leu (RON, 1 RON = 10,000 old lei), as in Chapter 0
 - ▶ **Low volatility** (4.4% per year) but **high excess kurtosis** (15.1)
 - ▶ Long calm periods interrupted by sudden adjustments
 - ▶ Largest daily moves: October 2008 (+2.9% on 3 October, −2.5% on 9 October)
- ▣ **Positive** lag-1 autocorrelation (0.17), unlike equities
 - ▶ The central bank smooths adjustments over several days; a daily fixing, not a traded price
- ▣ Lessons: facts are market-specific; low volatility is not low tail risk; exchange quote series showed spurious negative autocorrelation (quote errors)

Romania: The BET Total Return Index

- ▣ BET-TR, 2014–2026, 2,992 days
- ▣ Facts confirmed: excess kurtosis 17.8, clustering ($Q(10)$ on r_t^2 : 714), VaR 1% 2.7%
- ▣ Strongest negative skewness of the five series (-1.40)
 - ▶ Worst day (log return) -11.9% on 19 December 2018: reaction to the bank-asset tax announced by the government on 18 December
 - ▶ The tax was adopted on 28 December 2018 as Government Emergency Ordinance (GEO) 114/2018
 - ▶ Policy shocks produce single-day crashes without comparable rallies

Romania: Autocorrelation in a Smaller Market

- ▣ Positive lag-1 autocorrelation (0.06), in contrast with the S&P 500 (-0.08)
- ▣ **Non-synchronous trading:** index constituents do not all trade at every moment
 - ▶ The index reflects news with a delay, creating positive autocorrelation, [Campbell, Lo & MacKinlay \(1997\)](#), ch. 3
- ▣ Frontier and emerging markets (below developed status in index-provider classifications)
 - ▶ Lower liquidity (it is costlier to trade quickly)
 - ▶ Stronger autocorrelation and policy-driven jumps

Why Use Open, High, Low and Close?

- **Close-to-close** estimator: the standard deviation of daily close-to-close log returns
 - ▶ Uses one number per day and ignores intraday information
- **Range-based** estimators use the OHLC prices
 - ▶ Notation: o_t, h_t, l_t, c_t are the log open, high, low and close of day t
 - ▶ The range $h_t - l_t$ is large on volatile days even if the close equals the open
- **Efficiency**: the ratio of variances of two estimators; higher efficiency = less noise for the same sample

Parkinson and Garman–Klass Estimators

- ▣ **Parkinson**, [Parkinson \(1980\)](#): uses only the daily range

$$\hat{\sigma}_P^2 = \frac{1}{4 \ln 2} \frac{1}{n} \sum_{t=1}^n (h_t - l_t)^2$$

- ▣ **Garman–Klass**, [Garman & Klass \(1980\)](#): combines the range with the open-to-close move

$$\hat{\sigma}_{GK}^2 = \frac{1}{n} \sum_{t=1}^n \left[\frac{1}{2} (h_t - l_t)^2 - (2 \ln 2 - 1) (c_t - o_t)^2 \right]$$

- ▣ Derived for a Brownian motion without drift and without overnight jumps
 - ▶ Brownian motion: the continuous-time random walk (Chapter 11)
 - ▶ Efficiency relative to close-to-close: about 5 (Parkinson) and 7–8 (Garman–Klass)

Rogers–Satchell and Yang–Zhang Estimators

- **Rogers–Satchell**, Rogers & Satchell (1991): unbiased under a non-zero **drift** (expected trend of the log price)

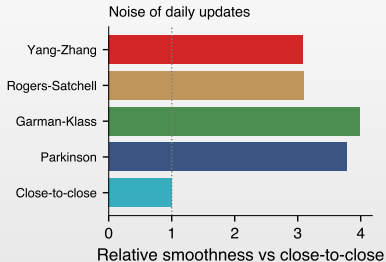
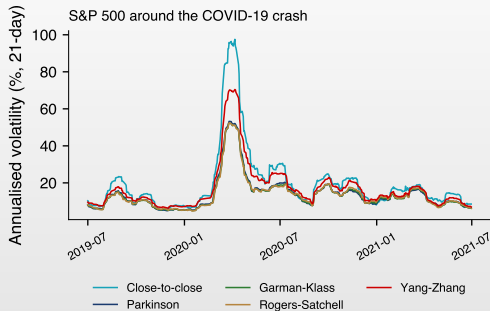
$$\hat{\sigma}_{RS}^2 = \frac{1}{n} \sum_{t=1}^n [(h_t - o_t)(h_t - c_t) + (l_t - o_t)(l_t - c_t)]$$

- **Yang–Zhang**, Yang & Zhang (2000): adds the overnight (close-to-open) jump

$$\hat{\sigma}_{YZ}^2 = \hat{\sigma}_{\text{overnight}}^2 + k \hat{\sigma}_{\text{open-close}}^2 + (1 - k) \hat{\sigma}_{RS}^2, \quad k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}$$

- ▶ $\hat{\sigma}_{\text{overnight}}^2$: variance of $o_t - c_{t-1}$; $\hat{\sigma}_{\text{open-close}}^2$: variance of $c_t - o_t$
 - ▶ Minimum-variance combination among drift-independent estimators robust to opening jumps
- Range estimators are the bridge to realised volatility from intraday data (Chapter 9)

Estimators on the S&P 500



- 21-day windows, 2008–2026; averages: close-to-close 16.4%, Yang–Zhang 13.9%, Parkinson 13.0%, Garman–Klass 12.4%, Rogers–Satchell 12.3%
- Pure range estimators are lower: they miss overnight gaps
- Yang–Zhang includes the overnight gap but stays 2.5 pp below close-to-close: the index open is partly stale (next slide)
- Range-based estimates are 3–4 times smoother (less noise in daily updates)

Data Quality: Stale Opens of the S&P 500 Index

- ▣ The index **open** is computed from the constituents' first prices; at 9:30 many stocks have not traded yet, so their previous close enters the open
- ▣ Open = previous close: about 96% of days until 2005, 3.5% over 2008–2026 (up to 26% in 2012–2013); on other days part of the open is still stale
- ▣ Comparison 2008–2026 with the SPY ETF, whose open is a traded price (annualised, %; the first two columns average 21-day estimates, the last two use the full sample)

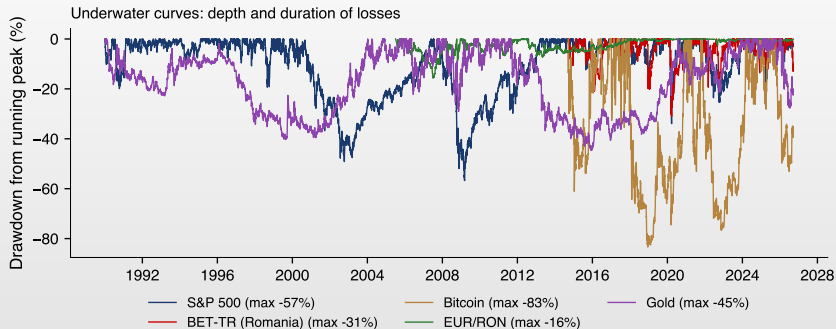
	Close-to-close	Yang–Zhang	Overnight	Open-to-close
S&P 500 index	16.4	13.9	6.7	17.3
SPY ETF	16.2	16.8	12.0	15.2

- ▣ A stale open moves part of the overnight move into the day: the index overnight volatility is understated and Yang–Zhang falls below close-to-close
- ▣ Lesson: for OHLC estimators use a traded instrument (ETF, futures), not the index

Drawdowns: Reminder and Focus

- Drawdown DD_t and maximum drawdown MDD were defined in Chapter 0
 - ▶ S&P 500: 9 October 2007 (1,565.15) to 9 March 2009 (676.53): -56.8% , which needs $+131\%$ to recover; peak regained on 28 March 2013
- New here: depth across five markets and **recovery duration**, the time from the peak until the price regains it
- MDD depends on the sample length: for a driftless Brownian motion of the log price, the maximum **log** drawdown $D = -\ln(1 + MDD) \geq 0$ has $\mathbb{E}[D] = \sqrt{\pi/2} \sigma \sqrt{T}$ Magdon-Ismail, Atiya, Pratap & Abu-Mostafa (2004)
 - ▶ $MDD = e^{-D} - 1$; drawdowns from samples of different length are not comparable (Seminar A4)
- Investors feel drawdowns more than volatility: a -50% loss needs $+100\%$ to recover

Drawdowns Across Markets



- Maximum drawdowns: Bitcoin -83%, S&P 500 -57%, gold -45%, BET-TR -31% (March 2020), EUR/RON -16%
- Depth and duration matter together

How Long Does Recovery Take?

	MDD	Peak	Trough	Peak regained
S&P 500	−57%	Oct 2007	Mar 2009	Mar 2013 (5.5 years)
BET-TR	−31%	Jan 2020	Mar 2020	Dec 2020 (0.9 years)
Bitcoin	−83%	Dec 2017	Dec 2018	Nov 2020 (3.0 years)
EUR/RON	−16%	Dec 2005	Jul 2007	Jan 2008 (2.0 years)
Gold	−45%	Sep 2011	Dec 2015	Jul 2020 (8.9 years)

- Depth and duration are different risks: gold's drawdown was shallower than Bitcoin's but lasted three times longer
- Fast V-shaped recoveries (BET-TR 2020) versus long underwater periods (gold 2011–2020)
- For EUR/RON a “drawdown” is a fall of the euro against the leu

Sharpe, Sortino and Calmar Ratios

- ▣ **Sharpe ratio** Sharpe (1966), Sharpe (1994): $SR = (\mu_{\text{ann}} - r_f) / \sigma_{\text{ann}}$ on **simple** returns, $\mu_{\text{ann}} = P\bar{R}$; $r_f = 0$ here, for comparability
 - ▶ S&P 500 1990–2026: $\mu_{\text{ann}} = 9.95\%$ (mean log return 8.33% plus $\approx \sigma^2/2 = 1.62$ pp), $\sigma_{\text{ann}} = 17.97\%$, $SR = 0.55$
- ▣ **Sortino ratio** Sortino & van der Meer (1991): $SoR = (\mu_{\text{ann}} - r_f) / \sigma_{\text{ann}}^-$, $\sigma^- = \sqrt{\frac{1}{T} \sum_{t=1}^T \min(R_t, 0)^2}$
 - ▶ The average runs over all T days, with positive days counted as 0; S&P 500: $\sigma_{\text{ann}}^- = 12.68\%$, $SoR = 0.78$
- ▣ **Calmar ratio** = $\text{CAGR} / |MDD|$; S&P 500: $8.69\% / 56.78\% = 0.15$, driven by one episode

Sharpe Ratios with Standard Errors

- Delta method, i.i.d. returns from the Normal distribution: $\text{Var}(\widehat{SR}_d) \approx (1 + SR_d^2/2)/T$ at the sampling frequency [Lo \(2002\)](#)
 - ▶ Annualised from T daily returns over Y years: $\text{SE}(\widehat{SR}_{\text{ann}}) \approx \sqrt{P(1 + SR_d^2/2)/T} \approx 1/\sqrt{Y}$: the span matters, not the frequency
 - ▶ i.i.d., any distribution with $\mathbb{E}r^4 < \infty$: $(1 - S SR_d + \frac{K-1}{4} SR_d^2)/T$ [Opdyke \(2007\)](#); here the SEs change by at most 0.02

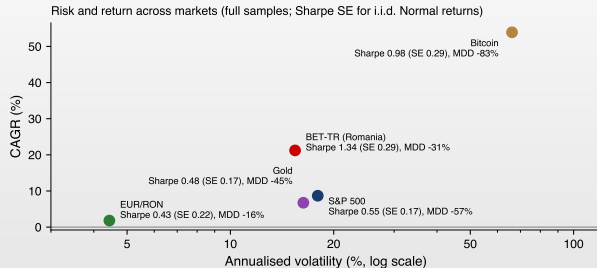
	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Years Y	36.7	12.0	12.0	21.2	36.7
Sharpe ratio (SE, Normal)	0.55 (0.17)	1.34 (0.29)	0.98 (0.29)	0.43 (0.22)	0.48 (0.17)

- Clustering: HAC or studentised block-bootstrap tests [Ledoit & Wolf \(2008\)](#); they still need finite fourth moments, which the Hill estimates put in doubt
 - ▶ BET-TR vs S&P 500, common days 2014–2026: difference 0.61, $z = 1.75$ ($p = 0.08$), percentile block-bootstrap 95% CI $[-0.01, 1.21]$ (Seminar B13)
- Calmar has no usable sampling theory: its drawdown grows with the sample length ($\sqrt{T}$ for a driftless log price)

Caveats for Performance Measures

- All three ratios are estimates with large sampling error: $SE(\widehat{SR}) \approx 1/\sqrt{Y}$
- Heavy tails and negative skewness make the Sharpe ratio flattering
 - ▶ The Deflated Sharpe Ratio corrects for this and for multiple testing (Chapter 13)
- Different sample periods make comparisons across markets fragile
- Currency and interest rates: BET-TR is in RON, where short-term rates were well above USD rates; with $r_f = 0$ its Sharpe ratio is inflated
- Always report the ratios together with drawdowns and tail statistics

Risk and Return Across Markets



	CAGR	Sharpe	Sortino	Calmar
S&P 500	8.7%	0.55	0.78	0.15
BET-TR	21.2%	1.34	1.87	0.68
Bitcoin	53.9%	0.98	1.45	0.65
EUR/RON	1.8%	0.43	0.65	0.12
Gold	6.7%	0.48	0.69	0.15

- Different sample periods: not a horse race
- S&P 500 without dividends; BET-TR with dividends
- Sharpe ratios on simple returns; SEs (i.i.d. Normal) in the chart labels

 MFM_ch1_risk_measures

Stylised Facts in One Table

Fact	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Heavy tails (Crow–Siddiqui > 0)	yes	yes	yes	yes	yes
Negative skewness	yes	strong	yes	no	yes
Volatility clustering (ARCH-LM)	yes	yes	yes	yes	yes
Slow decay of ACF $ r_t $	yes	weaker	weaker	yes	yes
Near-zero ACF of r_t	yes	small +	yes	no (+0.17)	yes
Leverage effect ($k = 1$)	−0.12	−0.12	−0.07	+0.10	−0.02

- Heavy tails, clustering and persistent volatility recur in all five samples; true long memory needs more evidence (breaks, Section 4)
- Autocorrelation and asymmetry depend on market microstructure and institutions

What a Good Model Must Reproduce

Requirements

- ▣ Time-varying, persistent volatility
- ▣ Heavy-tailed innovations, also after conditioning on volatility
- ▣ Asymmetric response to bad news
- ▣ Approximately uncorrelated returns

- ▣ The random walk with i.i.d. increments following the Normal distribution (Bachelier–Black–Scholes world) violates Facts 2, 3 and 5–11; Fact 4 holds trivially (sums of Normally distributed variables follow the Normal distribution)

Where in this course

- ▣ Ch. 5: GARCH; Ch. 6: multivariate volatility and dependence
- ▣ Ch. 7–8: VaR, Expected Shortfall, extreme value theory; backtesting
- ▣ Ch. 9: realised volatility from intraday data
- ▣ Ch. 13: machine learning must respect the low signal-to-noise ratio

Practical Checklist for Empirical Work

- ▣ **Know your data**
 - ▶ Price vs total-return index; adjusted vs raw prices; fixing vs traded price
 - ▶ Official reference rates where possible (e.g. EUR/RON)
- ▣ **Clean before you model**
 - ▶ Non-trading days (zero volume), holidays, stale quotes, obvious errors
 - ▶ Check extremes: are the largest moves real events?
- ▣ **Use log returns** for single-asset dynamics, simple returns for portfolios
- ▣ **Report robust statistics:** quantiles and tail counts next to kurtosis
- ▣ **Annualise carefully:** 252 vs 365 periods; $\sqrt{P}$ fails under dependence
- ▣ **Reproducibility:** code and data on GitHub, as Quantlets

Key Formulas

Returns

- ▣ $R_t = P_t/P_{t-1} - 1, \quad r_t = \ln(1 + R_t)$
- ▣ $r_t(k) = \sum_{j=0}^{k-1} r_{t-j}$
- ▣ $R_{p,t} = \sum_i w_i R_{i,t}$
- ▣ $\text{CAGR} = (P_T/P_0)^{1/Y} - 1$

Tests

- ▣ $JB = T[\hat{S}^2/6 + (\hat{K} - 3)^2/24]$
- ▣ Hill: $\text{SE}(\hat{\alpha}) \approx \hat{\alpha}/\sqrt{k}$
- ▣ $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k)$

Volatility

- ▣ $\sigma_{\text{ann}} = \sqrt{P} \hat{\sigma}$
- ▣ Parkinson: $\frac{1}{4 \ln 2} \overline{(h-l)^2}$
- ▣ Mixture: $K = 3 \mathbb{E}[\sigma^4] / \mathbb{E}[\sigma^2]^2$

Performance

- ▣ $SR = (\mu - r_f)/\sigma$ on simple returns; $\text{SE} \approx 1/\sqrt{Y}$
- ▣ $SoR = (\mu - r_f)/\sigma^-$
- ▣ $DD_t = P_t / \max_{s \leq t} P_s - 1$
- ▣ Calmar = $\text{CAGR} / |\text{MDD}|$

An Open Question: Are Tail Indices Stable over Time?

- The stylised facts are usually stated for a whole sample; the tail index α is treated as a constant
- Crypto-assets changed in structure: futures (2017), spot Bitcoin ETFs (January 2024), institutional holders
 - ▶ [Pele et al. \(2023\)](#): the tail factor is what separates cryptos from other asset classes
- Question: did institutionalisation make Bitcoin's tails thinner, i.e. raise α ?
- Inference is delicate: $\hat{\alpha}$ uses only the k largest observations, and returns are dependent
 - ▶ [Hoga \(2017\)](#): change-point tests for the tail index valid under β -mixing (e.g. GARCH-type) dependence

The Discovery Loop, Step by Step (1/2)

□ Literature

- ▶ Find peer-reviewed work since 2015 on time variation or structural breaks in tail indices of asset returns, with tests valid under GARCH-type dependence. For each: DOI, estimator, test, data. Flag entries you are unsure of.
- ▶ Resolve every DOI on Crossref; the title and authors must match

□ Hypothesis

- ▶ State competing hypotheses for a change in Bitcoin's tail after the 2024 spot ETFs: a change in tail shape, a fall in volatility only, sampling noise. What does each predict for the Hill index of raw returns and of GARCH-standardised returns?

□ Data and code

- ▶ Read BTC-USD.CC from data/market (close); log returns on its own calendar; Hill index of $|r|$ with $k = 2.5\%$ of n before and after 11 January 2024; moving-block bootstrap standard errors.
- ▶ Replicate a known number first: the chapter's Hill estimates, Bitcoin 4.2 and S&P 500 3.1

The Discovery Loop, Step by Step (2/2)

▣ Robustness

- ▶ Write a pre-registered robustness plan: k from 1% to 5% of n , loss and gain tails separately, GARCH-standardised residuals, block lengths from 10 to 50 days, a placebo date and a placebo asset.

▣ Critique

- ▶ Referee this design. With 24 tail observations after the break, what is the power against a change of 1 in the tail index? Which dependence assumption does each standard error need?

▣ Report

- ▶ $\hat{\alpha}$ in each window, both standard errors, the placebo, the minimum detectable change
- ▶ Prompts that shaped a decision go into AI_USE.md

- ▣ Scale versus shape: a lower volatility leaves α unchanged; only a change in the shape of the tail moves it

What the Researcher Must Check

- **Wrong return or price series:** prices, simple returns or a different calendar
 - ▶ test: reproduce the chapter's Hill table before anything else
- **Choice of k after seeing the result:** fix the rule ($k = 2.5\%$ of n) in advance; show the Hill plot for 1–5%
- **i.i.d. standard errors:** $\hat{\alpha}/\sqrt{k}$ ignores volatility clustering
 - ▶ test: compare with a block bootstrap; for the S&P 500 before 2024 it is 2.2 times larger
- **Break date chosen from the data:** use a date fixed by the event, or a sup-type change-point test with its own critical values ([Hoga \(2017\)](#))
- **Bad ticks create fake tails:** list the ten largest $|r_t|$ and match each to a dated event
- **Low power:** report the minimum detectable change: 2.8 standard errors of the difference for a two-sided 5% test with 80% power, 2.5 for a one-sided test

Mini-Case: Did Bitcoin's Tail Get Thinner after the ETFs?

- A typical assistant proposal: “institutional money since the spot ETFs has tamed Bitcoin's tails”
- Test: Hill index of $|r_t|$, $k = 2.5\%$ of n ; $H_0 : \alpha_{\text{post}} = \alpha_{\text{pre}}$ against $H_1 : \alpha_{\text{post}} > \alpha_{\text{pre}}$
 - ▶ $z = (\hat{\alpha}_{\text{post}} - \hat{\alpha}_{\text{pre}}) / \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$, s from a moving-block bootstrap (blocks of 20 days)
- Before (Sep 2014 – 10 Jan 2024, $k = 85$): $\hat{\alpha} = 4.16$ ($s = 0.52$)
- After (11 Jan 2024 – Sep 2026, $k = 24$): $\hat{\alpha} = 4.31$ ($s = 0.91$); $z = 0.15$
- Placebo, S&P 500 at the same date: $2.82 \rightarrow 2.86$, $z = 0.03$
- Interpretation: no evidence of a change, but for this one-sided 5% test with 80% power the minimum detectable change is $(1.645 + 0.842) \times 1.05 = 2.6$
 - ▶ a normal approximation, rough with 24 tail observations: the data cannot distinguish “unchanged” from a large thinning of the tail; wait for more data or pool assets

 ai_discovery_case.py

Your Turn: A Project Seed

- **Question:** did the tail index of Bitcoin and Ethereum change with institutionalisation (futures 2017, spot ETFs 2024)?
 - ▶ raw and GARCH-standardised returns: shape versus scale; loss and gain tails separately
- **Replicate first:** the chapter's Hill table ($k = 2.5\%$ of n) and the mini-case numbers
- **Deliverables**
 - ▶ a pre-registration note (H_0 , H_1 , rule for k , break dates, bootstrap) committed before the results
 - ▶ the change-point test of [Hoga \(2017\)](#) as a Quantlet; a report with power calculations
- **Logs:** prompts in `AI_USE.md`; every caught error (an i.i.d. standard error, a date chosen from the data) in `AI_ERRORS.md`

Conclusions

- ▣ Returns are **uncorrelated but not independent**: their size is predictable, their linear autocorrelation is weak
- ▣ Heavy tails are the rule: 5σ days are 1000+ times more frequent than the Normal distribution predicts
- ▣ Volatility clusters, is highly persistent (long memory or breaks) and reacts asymmetrically to losses in equity markets
- ▣ Tails thin out with aggregation, but slowly
- ▣ Crypto shows the same facts with larger amplitude; managed exchange rates and thin markets add autocorrelation
- ▣ Range-based estimators extract more information from OHLC data
- ▣ Risk-adjusted measures must be read together with drawdowns and tail statistics

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Statistics of Financial Markets
- ▣  Tukey's g- and h- transformations
- ▣  Applied Time Series Analysis with Python

Code and data

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_01

References (1/4)

- Andersen, T.G., Bollerslev, T., Diebold, F.X., Ebens, H. (2001). The distribution of realized stock return volatility. *Journal of Financial Economics*, 61(1), 43–76.
- Bachelier, L. (1900). *Théorie de la spéculation*. *Annales scientifiques de l'École Normale Supérieure*, 17, 21–86.
- Bai, J., Ng, S. (2005). Tests for skewness, kurtosis, and normality for time series data. *Journal of Business & Economic Statistics*, 23(1), 49–60.
- Bouchaud, J.-P., Matacz, A., Potters, M. (2001). Leverage effect in financial markets: the retarded volatility model. *Physical Review Letters*, 87(22), 228701.
- Campbell, J.Y., Lo, A.W., MacKinlay, A.C. (1997). *The Econometrics of Financial Markets*. Princeton University Press.
- Christie, A.A. (1982). The stochastic behavior of common stock variances: value, leverage and interest rate effects. *Journal of Financial Economics*, 10(4), 407–432.
- Clark, P.K. (1973). A subordinated stochastic process model with finite variance for speculative prices. *Econometrica*, 41(1), 135–155.
- Cong, L.W., Li, X., Tang, K., Yang, Y. (2023). Crypto wash trading. *Management Science*, 69(11), 6427–6454.
- Cont, R. (2001). Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2), 223–236.
- Dalla, V., Giraitis, L., Phillips, P.C.B. (2022). Robust tests for white noise and cross-correlation. *Econometric Theory*, 38(5), 913–941.
- Davis, R.A., Mikosch, T. (1998). The sample autocorrelations of heavy-tailed processes with applications to ARCH. *Annals of Statistics*, 26(5), 2049–2080.
- Diebold, F.X., Inoue, A. (2001). Long memory and regime switching. *Journal of Econometrics*, 105(1), 131–159.
- Ding, Z., Granger, C.W.J., Engle, R.F. (1993). A long memory property of stock market returns and a new model. *Journal of Empirical Finance*, 1(1), 83–106.

References (2/4)

- Drees, H., Resnick, S., de Haan, L. (2000). [How to make a Hill plot](#). *Annals of Statistics*, 28(1), 254–274.
- Engle, R.F. (1982). [Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation](#). *Econometrica*, 50(4), 987–1007.
- Fama, E.F. (1965). [The behavior of stock-market prices](#). *Journal of Business*, 38(1), 34–105.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- Gabaix, X., Ibragimov, R. (2011). [Rank – 1/2: a simple way to improve the OLS estimation of tail exponents](#). *Journal of Business & Economic Statistics*, 29(1), 24–39.
- Garman, M.B., Klass, M.J. (1980). [On the estimation of security price volatilities from historical data](#). *Journal of Business*, 53(1), 67–78.
- Geweke, J., Porter-Hudak, S. (1983). [The estimation and application of long memory time series models](#). *Journal of Time Series Analysis*, 4(4), 221–238.
- Granger, C.W.J., Ding, Z. (1995). [Some properties of absolute return: an alternative measure of risk](#). *Annales d'Économie et de Statistique*, 40, 67–91.
- Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). [Applied Quantitative Finance](#), 3rd ed. Springer.
- Härdle, W.K., Harvey, C.R., Reule, R.C.G. (2020). [Understanding cryptocurrencies](#). *Journal of Financial Econometrics*, 18(2), 181–208.
- Hill, B.M. (1975). [A simple general approach to inference about the tail of a distribution](#). *Annals of Statistics*, 3(5), 1163–1174.
- Hoga, Y. (2017). [Change point tests for the tail index of \$\beta\$ -mixing random variables](#). *Econometric Theory*, 33(4), 915–954.
- Hsing, T. (1991). [On tail index estimation using dependent data](#). *Annals of Statistics*, 19(3), 1547–1569.

References (3/4)

- Inclán, C., Tiao, G.C. (1994). Use of cumulative sums of squares for retrospective detection of changes of variance. *Journal of the American Statistical Association*, 89(427), 913–923.
- Jarque, C.M., Bera, A.K. (1980). Efficient tests for normality, homoscedasticity and serial independence of regression residuals. *Economics Letters*, 6(3), 255–259.
- Karpoff, J.M. (1987). The relation between price changes and trading volume: a survey. *Journal of Financial and Quantitative Analysis*, 22(1), 109–126.
- Kim, T.-H., White, H. (2004). On more robust estimation of skewness and kurtosis. *Finance Research Letters*, 1(1), 56–73.
- Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- Ljung, G.M., Box, G.E.P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.
- Lo, A.W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- Lobato, I., Nankervis, J.C., Savin, N.E. (2001). Testing for autocorrelation using a modified Box–Pierce Q test. *International Economic Review*, 42(1), 187–205.
- Magdon-Ismail, M., Atiya, A.F., Pratap, A., Abu-Mostafa, Y.S. (2004). On the maximum drawdown of a Brownian motion. *Journal of Applied Probability*, 41(1), 147–161.
- Mandelbrot, B. (1963). The variation of certain speculative prices. *Journal of Business*, 36(4), 394–419.
- Mikosch, T., Stărică, C. (2004). Nonstationarities in financial time series, the long-range dependence, and the IGARCH effects. *Review of Economics and Statistics*, 86(1), 378–390.
- Opdyke, J.D. (2007). Comparing Sharpe ratios: so where are the p-values?. *Journal of Asset Management*, 8(5), 308–336.
- Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1), 61–65.
- Pele, D.T. (2014). A SAS approach for estimating the parameters of an alpha-stable distribution. *Procedia Economics and Finance*, 10, 68–77.

References (4/4)

- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). [Information entropy and measures of market risk](#). *Entropy*, 19(5), 226.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). [Are cryptos becoming alternative assets?](#). *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Praetz, P.D. (1972). [The distribution of share price changes](#). *Journal of Business*, 45(1), 49–55.
- ▣ Ratliff-Crain, E., Van Oort, C.M., Koehler, M., Tivnan, B.F. (2025). [Revisiting Cont's stylized facts for modern stock markets](#). *Quantitative Finance*, 25(9), 1343–1373.
- ▣ Robinson, P.M. (1995). [Gaussian semiparametric estimation of long range dependence](#). *Annals of Statistics*, 23(5), 1630–1661.
- ▣ Rogers, L.C.G., Satchell, S.E. (1991). [Estimating variance from high, low and closing prices](#). *Annals of Applied Probability*, 1(4), 504–512.
- ▣ Romano, J.P., Thombs, L.A. (1996). [Inference for autocorrelations under weak assumptions](#). *Journal of the American Statistical Association*, 91(434), 590–600.
- ▣ Sharpe, W.F. (1966). [Mutual fund performance](#). *Journal of Business*, 39(1), 119–138.
- ▣ Sharpe, W.F. (1994). [The Sharpe ratio](#). *Journal of Portfolio Management*, 21(1), 49–58.
- ▣ Sortino, F.A., van der Meer, R. (1991). [Downside risk](#). *Journal of Portfolio Management*, 17(4), 27–31.
- ▣ Student (1908). [The probable error of a mean](#). *Biometrika*, 6(1), 1–25.
- ▣ Taylor, S.J. (1986). *Modelling Financial Time Series*. Wiley; 2nd ed. (2008), World Scientific, doi:10.1142/6578.
- ▣ Yang, D., Zhang, Q. (2000). [Drift-independent volatility estimation based on high, low, open, and close prices](#). *Journal of Business*, 73(3), 477–492.